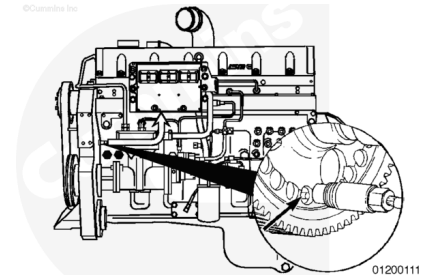


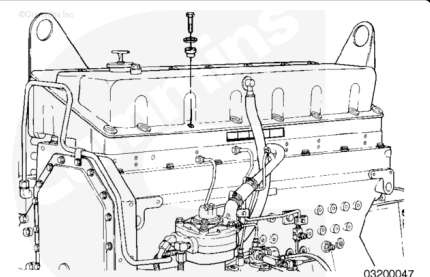
Trainings ▾

Measure

A condition on M series engines where the engine will **not** start or is difficult to start can be caused by excessive clearance between the engine position sensor and the pickup on the back of the camshaft gear. This condition prevents the engine position sensor from sending the necessary signal to the electronic control module. The excessive clearance can be caused by a failed thrust bearing on the camshaft. Use this procedure to measure the camshaft end clearance in-chassis on M series engines without removing the front gear cover. If the end clearance is **not** within the specifications given, the gear cover **must** be removed and the cause of the excessive end clearance repaired.

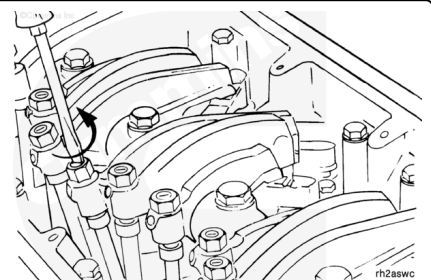


To measure the camshaft end clearance without removing the front gear cover, first remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)



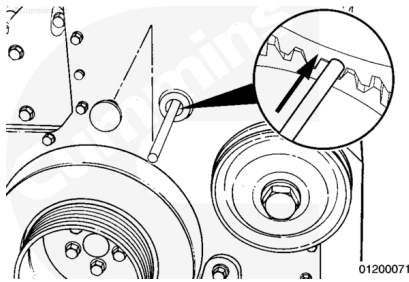
Loosen the locknut and turn out the adjusting screw on each injector and valve rocker lever. The overhead load on the camshaft **must** be released to allow free movement of the camshaft.

It is **not** necessary to remove the push rods or push tubes.



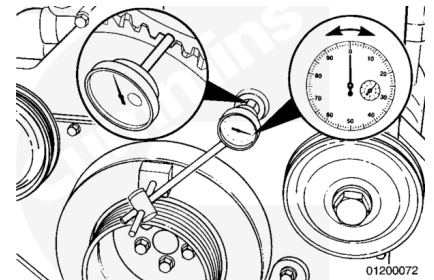
Remove the inspection plug in the front of the gear cover.

Use a rod against the camshaft gear to push the camshaft as far as possible toward the rear of the engine.



Set up a dial indicator gauge, Part Number 3376050, and extension, Part Number ST-537-4, with a magnetic base on the front of the cylinder head or vibration damper. Place the tip of the indicator on the face of the camshaft gear through the gear cover inspection hole.

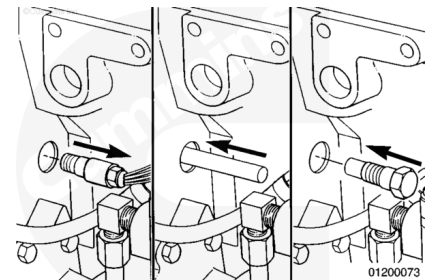
Set the dial indicator to zero.



Remove the engine position sensor from the rear of the gear housing.

Use a rod to reach inside the gear housing through the engine position sensor mounting hole and push the camshaft gear as far as possible toward the front of the engine.

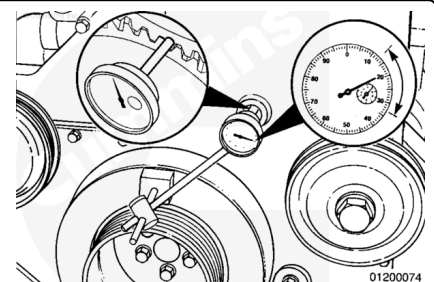
If access limits make it difficult to push the camshaft by hand, a threaded bolt (3/4 x 16) can be threaded into the hole and used as a pushing device against the back of the cam gear. The capscrew needs to have the threads ground off, approximately 25.4 mm [1 in] back from the end. Although the hole is threaded clear through the gear housing, this will make sure the capscrew does **not** contact anything but the camshaft gear.



Record the measurement on the dial indicator.

Camshaft End Clearance		
mm		in
0.13	MIN	0.005
0.41	MAX	0.016

If the camshaft end clearance does **not** meet the above specifications, the front cover **must** be removed and the camshaft thrust plate inspected. Refer to Procedure 001-008 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-008-tr.html)



Adjust the valve and injectors. Refer to Procedure 003-004 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-004-tr.html>)

Install the camshaft inspection plug in the gear cover.

Install the engine position sensor.

